

AZURE COST MANAGEMENT

Budget & Anomaly Alerting

Mini Project Report

Project: 24CC3046-P016
Domain: Cloud Computing
Platform: Microsoft Azure for Students

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Prepared from the live Azure Portal configuration and project screenshots.

1. Problem Statement

Cloud resources can continue consuming budget without clear ownership or timely visibility. For a student Azure subscription, the project focuses on giving the team a simple way to monitor spending, notify owners when a budget threshold is reached, and review cost anomalies.

Project Objectives

- Create a small Azure resource environment suitable for a student project.
- Apply ownership and project tags to improve accountability.
- Create a monthly subscription budget with threshold-based email alerts.
- Use Cost Analysis to review cost by service and identify major cost drivers when data is available.
- Configure an anomaly alert rule and route notifications to project owners.

Use Cases

Use Case 1: Alert department owners when spending approaches or exceeds the approved budget.

Use Case 2: Review monthly service costs and identify the top cost drivers for investigation.

Use Case 3: Detect unusual cost behaviour through an anomaly alert rule.

Bottlenecks and Challenges

Alerts can arrive after spend has occurred, so the project is designed for visibility and response rather than automatic prevention. Untagged resources can also make ownership and chargeback incomplete.

2. Azure Resources and Ownership

The implementation uses a dedicated resource group named RG-CostManagement and a storage account costalertstorage001. The storage account is deployed in East Asia and uses StorageV2, Standard performance, and locally redundant storage (LRS).

- Subscription: Azure for Students
- Resource group: RG-CostManagement
- Storage account: costalertstorage001
- Storage type: StorageV2 (general purpose v2)
- Performance: Standard
- Replication: Locally redundant storage (LRS)
- Location: East Asia

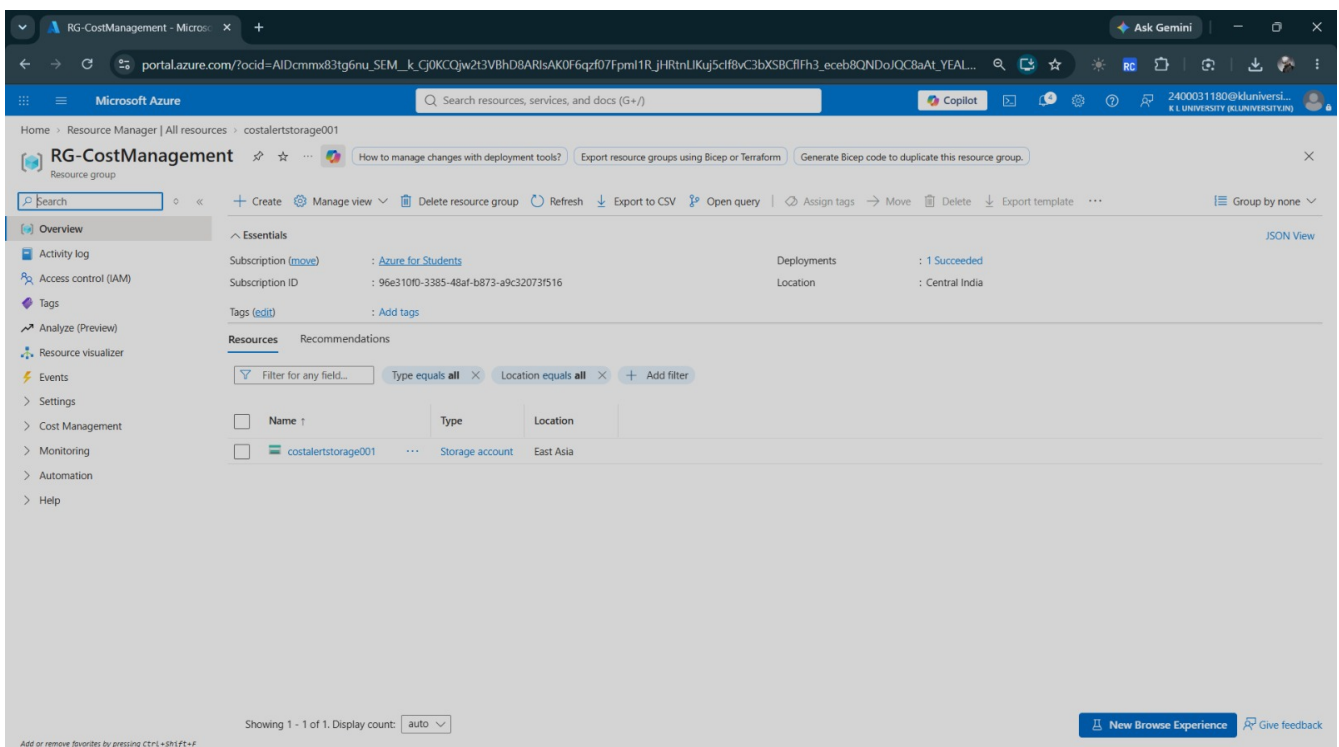


Figure 1. Azure resource group showing the deployed storage resource.

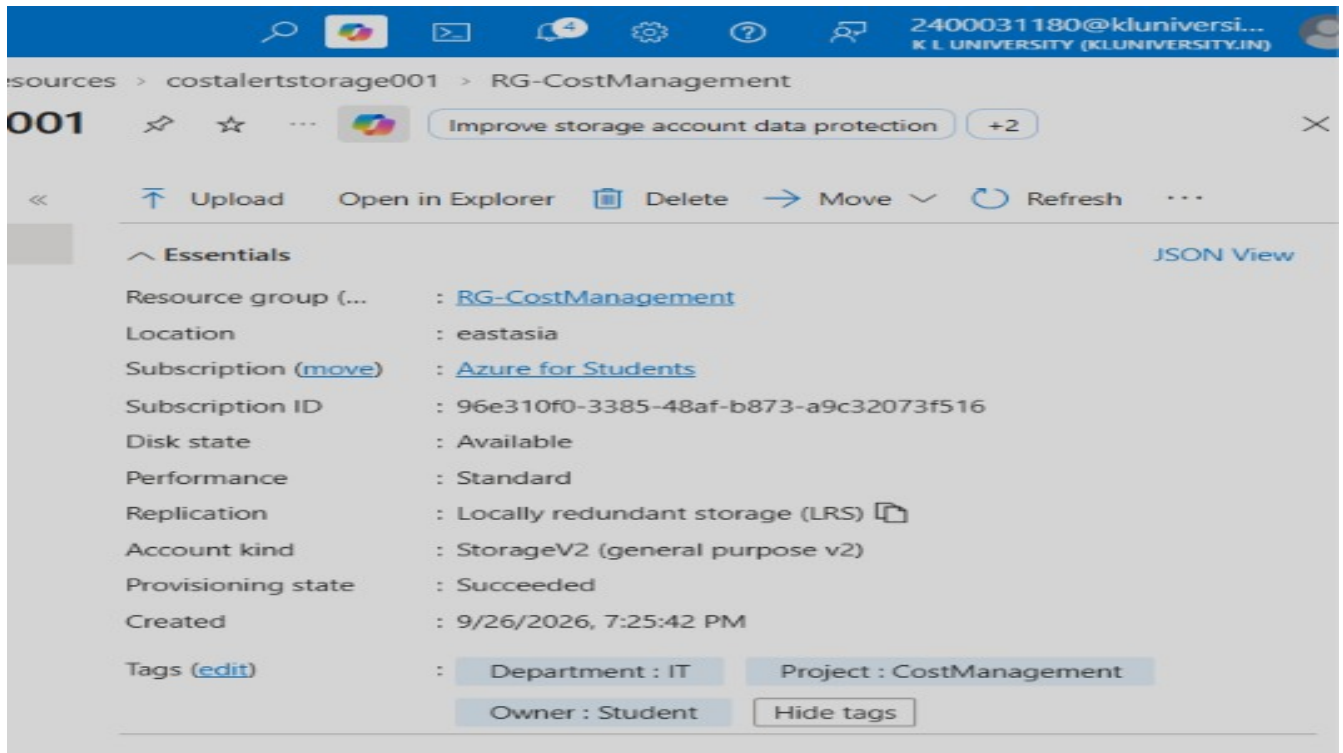


Figure 2. Storage account overview with project ownership tags: Department = IT, Project = CostManagement, Owner = Student.

3. Budget Configuration

A subscription-level monthly budget named MonthlyCostBudget was created for the Azure for Students subscription. The configured amount is ₹10 per month. Two actual-cost thresholds are configured for notification.

Setting	Configured value
Budget name	MonthlyCostBudget
Scope	Azure for Students (Subscription)
Budget amount	₹10.00
Period	Monthly
80% alert	₹8.00
100% alert	₹10.00
Alert recipients	2400031197@gmail.com; 2400031180@gmail.com

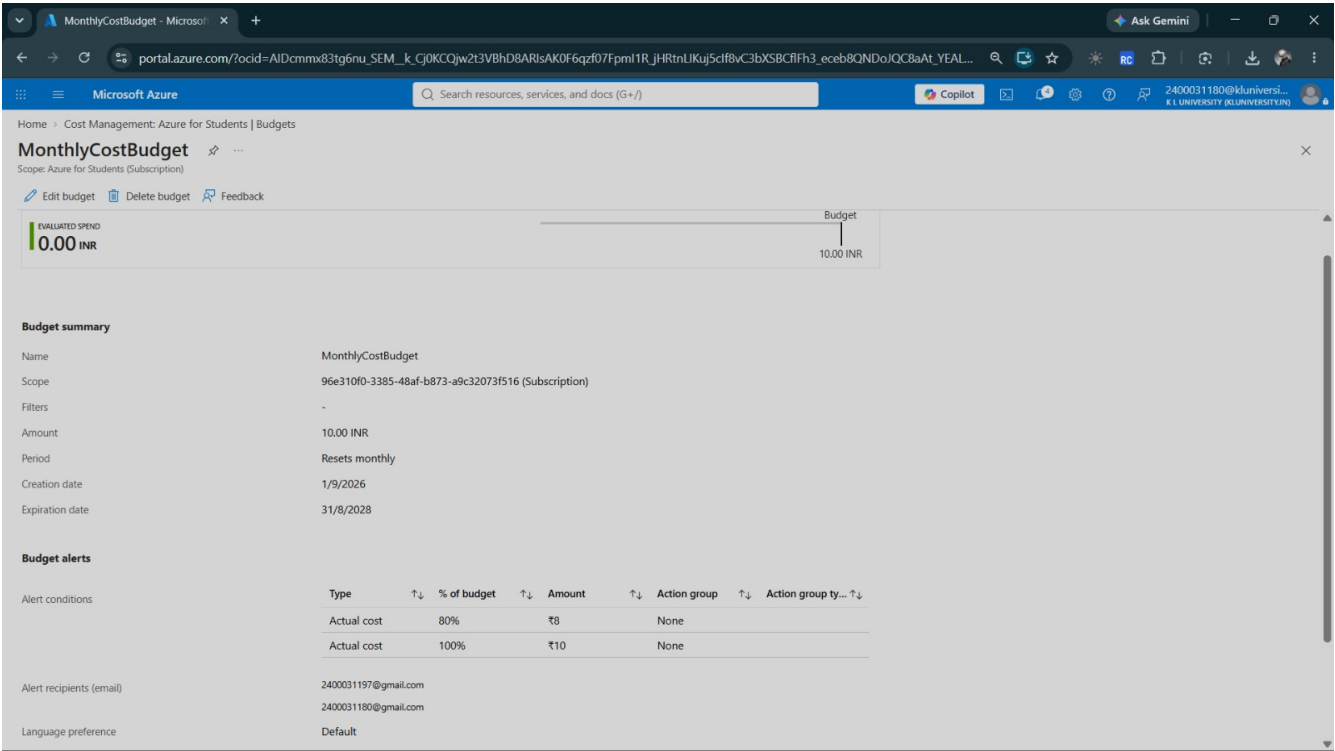


Figure 3. Live MonthlyCostBudget configuration and actual-cost alert thresholds.

The budget is a monitoring mechanism. It sends notifications when configured thresholds are reached; it does not automatically stop Azure resources.

4. Cost Analysis

A Cost Analysis view named Monthly Cost by Service is used to inspect the subscription by service name with monthly granularity. The current September 2026 view reports no cost data for the period.

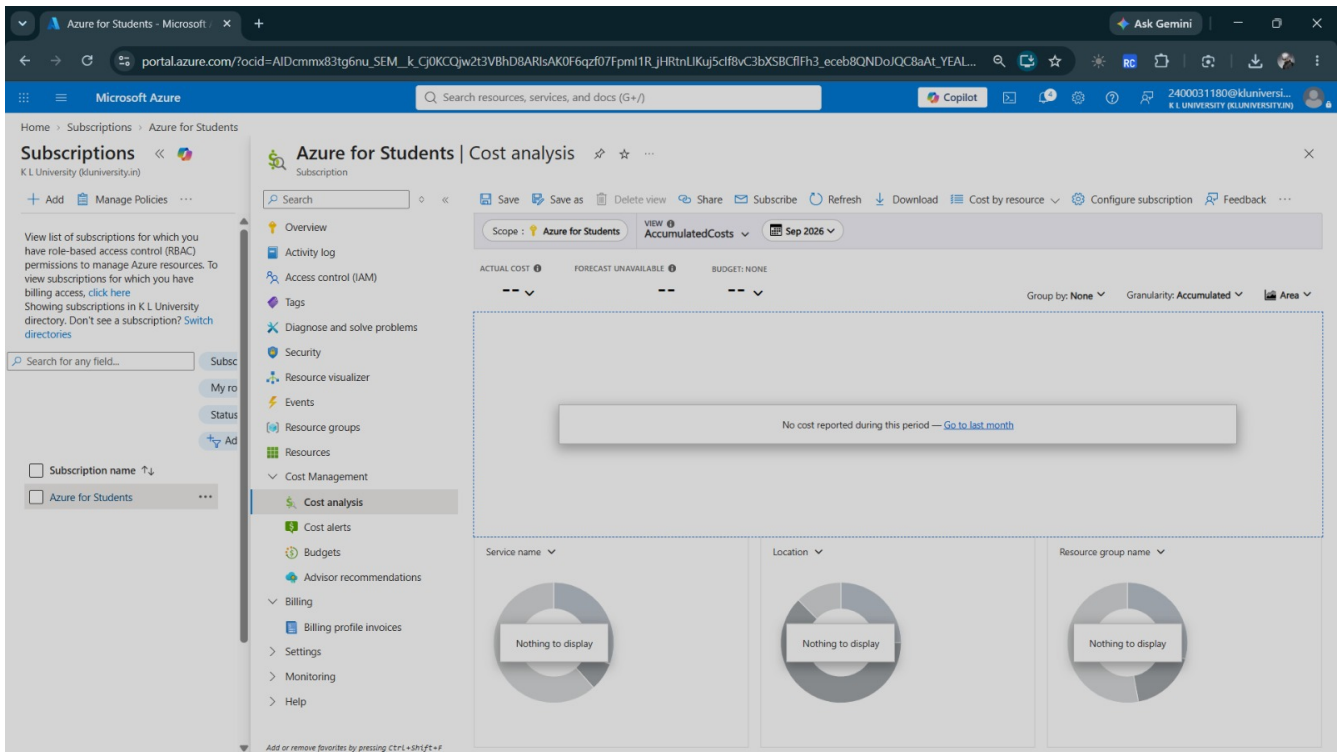


Figure 4. Cost Analysis view for September 2026.

Current observation

No cost reported during this period is shown by Azure. Therefore, no genuine top-five cost-driver values are available yet. The project does not invent or estimate those values; the Cost Analysis view is ready for use when billable cost data appears.

Intended monthly analysis

When cost data becomes available, the team can group and compare services, investigate unusual changes, and document the five highest-cost services for the month.

5. Anomaly Detection and Alerting

The project includes an active anomaly alert rule at the Azure for Students subscription scope. The rule is named Daily anomaly by resource group and is configured to notify two recipients.

Setting	Configured value
Alert type	Anomaly
View	Daily anomaly by resource group
Status	Active
Condition	Anomaly detected
Action	Email me and 1 other
Scope	Azure for Students
Recipients	2400031180@kluniversity.in; 2400031197@gmail.com

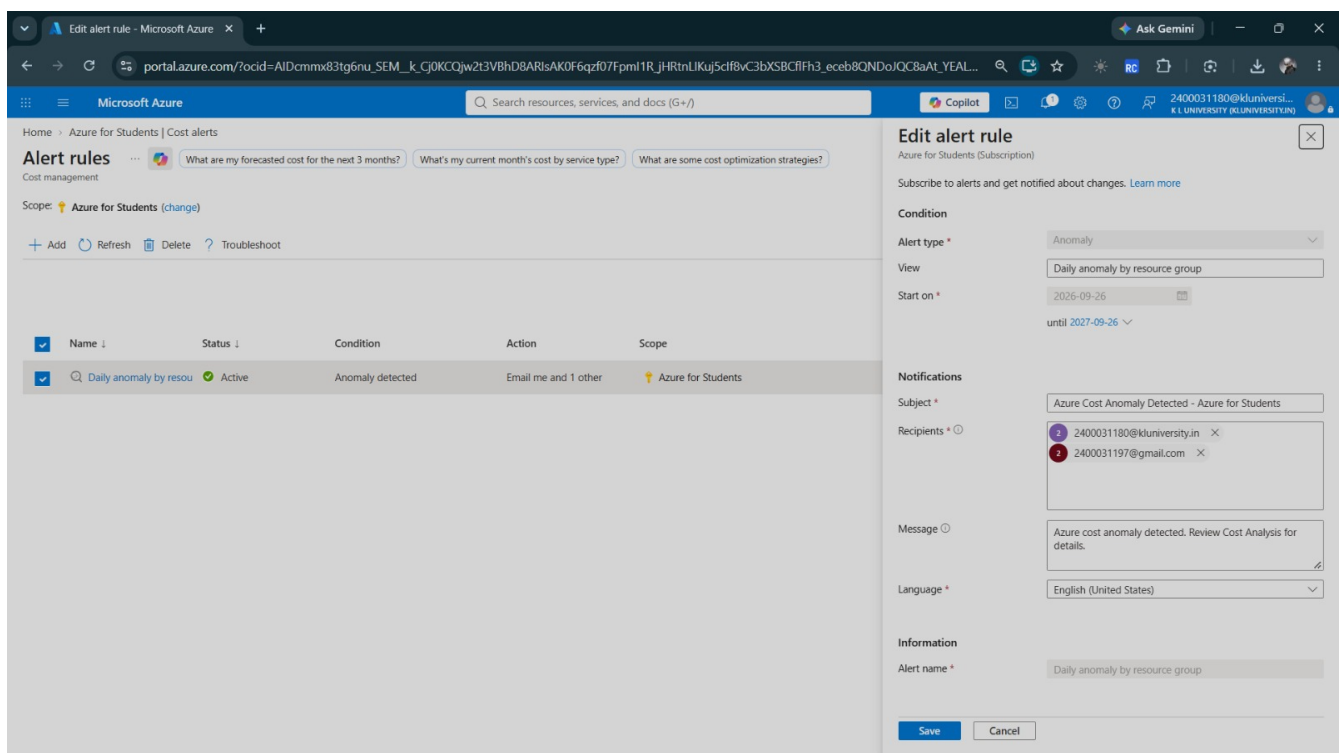


Figure 5. Active anomaly alert rule configured in Azure Cost Management.

The Cost alerts landing page may display “No Alerts to display” when no anomaly has actually triggered. The evidence for the configured control is the active rule shown above; it should not be presented as proof that an anomaly has already occurred.

6. Architecture and Workflow

The architecture connects Azure resource usage to Cost Management, where cost analysis, budget thresholds, and anomaly detection provide monitoring signals. Email notifications are then sent to the project owners, who investigate and take action.

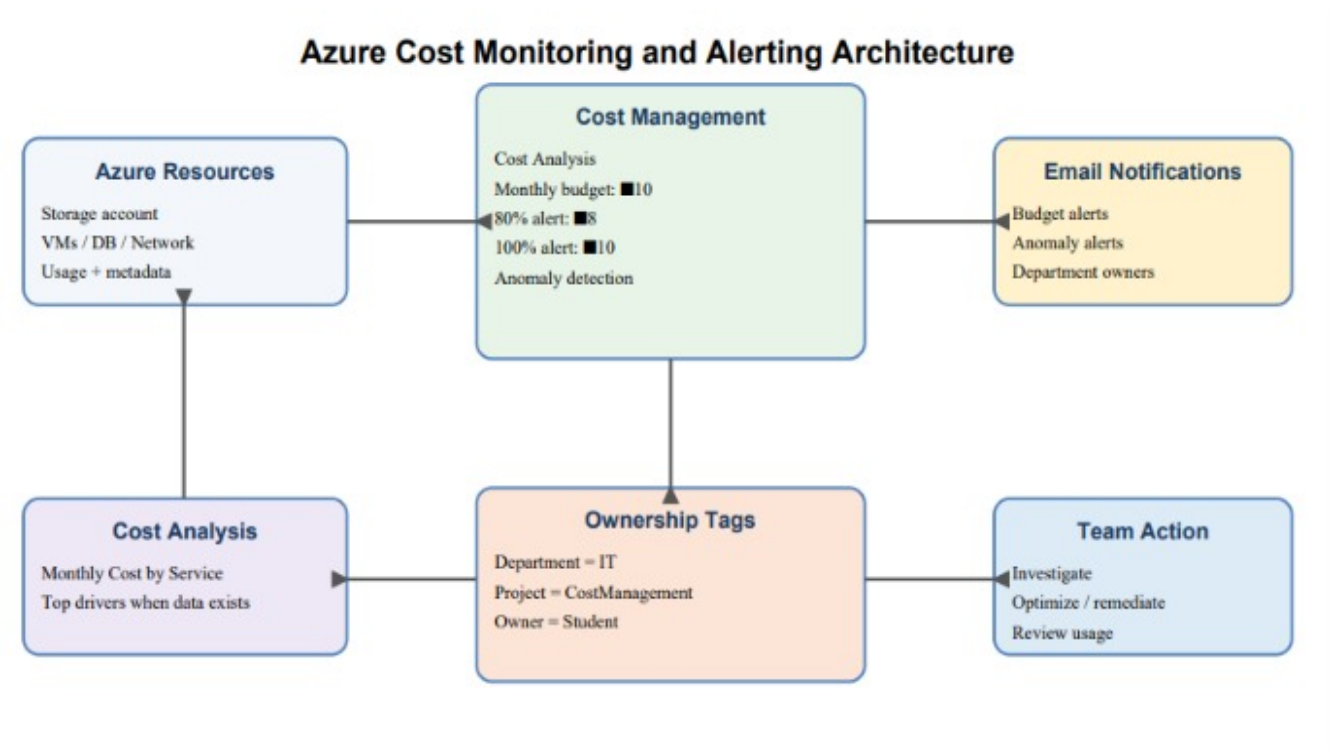


Figure 6. Azure Cost Monitoring and Alerting Architecture.

Stage	Project flow
1	Azure resources generate usage and cost data.
2	Cost Management collects and presents cost information.
3	Monthly budget evaluates actual spend against ₹8 and ₹10 thresholds.
4	Anomaly detection identifies unusual cost behaviour.
5	Email notifications inform the responsible owners.
6	The team reviews Cost Analysis and investigates the resource/service.

7. Implementation, Outcome and Reflection

Implementation Summary

- Created RG-CostManagement and the costalertstorage001 storage account.
- Applied Department, Project, and Owner tags for ownership tracking.
- Created the MonthlyCostBudget with a ₹10 monthly limit.
- Configured 80% (₹8) and 100% (₹10) actual-cost alerts.
- Prepared the Monthly Cost by Service Cost Analysis view.
- Created and activated the Daily anomaly by resource group alert rule.

Outcome

The project demonstrates the monitoring and alerting controls required for the use cases. The live Azure configuration provides evidence of resource organization, ownership tagging, budget thresholds, cost analysis, and anomaly alerting.

Bottlenecks / Limitations

- No cost has yet been reported in the current September 2026 Cost Analysis period, so actual monthly top-five values cannot be shown yet.
- Budget alerts notify the owners but do not automatically stop resources.
- Anomaly alerts depend on Azure detecting an anomaly; configuring a rule does not mean an anomaly has already occurred.
- Untagged resources would reduce the completeness of ownership or chargeback analysis.

Conclusion

Azure Cost Management provides a practical way to monitor student-cloud spending through budgets, cost analysis, ownership tags, and anomaly alerts. The current implementation establishes the required controls without fabricating cost results that are not yet present in the subscription.

Thank You